

Computational Frameworks

Streaming (Part 2)

OUTLINE

① Introduction to the streaming model (Part 1)

- Warm-up: finding the majority element

② Sampling (Part 2)

- Uniform sampling with reservoir sampling
- Finding frequent items with sticky sampling

③ Sketching (Part 3)

- Counting distinct elements with probabilistic counting
- Estimating moments with AMS sketch

Sampling

Sampling

Problem

Maintaining an m -sample S of a stream $\Sigma = x_1, x_2, \dots, x_n$.

- m -sample: a random sample S of m elements where each entry $x \in \Sigma$ has probability m/n to be in S .  Uniform sample
- If the size n of the stream is known:
 - Before the stream starts, compute an m -sample $\mathcal{I}$ of the set of integers $\{1, \dots, n\}$.
 - For each entry $x_i \in \Sigma$ with $i \in \mathcal{I}$, add x_i to S .

Sampling

The previous approach has some limitations:

- Stream size n must be known: however, in many cases, n is unknown (e.g., packets in a router).
- It cannot provide m -samples of prefixes if the stream. Specifically, for any $t < n$, the set $S \cap \{x_1, \dots, x_t\}$ is not an m -sample of $\{x_1, \dots, x_t\}$.

Remark: In some settings, streams are unbounded and we would like a continuous monitoring (e.g., sensor data)

Reservoir Sampling

The previous limitations are addressed in the **Reservoir Sampling** algorithm (introduced in 1985 by **J. Vitter**)

$$\Sigma \equiv x_1 x_2 \dots x_t \dots$$

For any element $x_t \in \Sigma$:

- If $t \leq m$, add x_t to S ;
- If $t > m$, with probability m/t do the following
 - Evict a random element x from S ;
 - Add x_t to S .

Insert the first m elements of Σ in S

Obs. At any time t $|S|=m$

Remark: the stream size needs not be known, and, in fact, it can be unbounded.

Reservoir Sampling

Theorem

Let $\Sigma = x_1, x_2, \dots$. For any time $t \geq m$, S is an m -sample of $x_1, x_2, \dots, x_t$.

let $S_t = S$ after processing x_t
We now show by induction on $t \geq m$ that
for any $i \in [1, t]$ $\text{Prob}(x_i \in S) = \frac{m}{t}$

$\Rightarrow S_t$ is an m -sample of $x_1, x_2, \dots, x_t$

Base of induction: $t = m$ (trivial)

Induction step: we show that the property holds
for $t > m$ assuming that it holds for $t-1$

Reservoir Sampling

H_p.

$$\text{Prob}(x_i \in S_{t-1}) = m/(t-1) \quad \forall i \in [1, t-1]$$

We now show the property for t

* $\text{Prob}(x_t \in S_t) = m/t$ by construction

* Consider $i < t$. Define event $A = "x_t \text{ is added to } S_t"$

$$\text{Prob}(x_i \in S_t) = \text{(by LAW of TOTAL PROBABILITIES)}$$

$$\text{Prob}(x_i \in S_t \wedge A) + \text{Prob}(x_i \in S_t \wedge \sim A)$$

$$\text{Prob}(x_i \in S_t \wedge A) = \text{Prob}(x_i \in S_t | A) * \text{Prob}(A)$$

$$\text{Prob}(x_i \in S_t \wedge \sim A) = \text{Prob}(x_i \in S_t | \sim A) * \text{Prob}(\sim A)$$

$$- \text{Prob}(x_i \in S_t | \sim A) * \text{Prob}(\sim A) = \underline{\frac{m}{t-1} \left(1 - \frac{m}{t}\right)}$$

Since:

$$- \text{Prob}(x_i \in S_t | \sim A) = \text{Prob}(x_i \in S_{t-1}) = \frac{m}{t-1} \quad \text{by ind. Hp.}$$

$$\text{Prob}(\sim A) = 1 - m/t$$

$$\text{Prob}(x_i \in S_t | A) * \text{Prob}(A) = \underline{\frac{m}{t-1} \left(1 - \frac{1}{m}\right) \cdot \frac{m}{t}}$$

Since:

$$\begin{aligned} \text{Prob}(x_i \in S_t | A) &= \text{Prob}(x_i \in S_{t-1}) * \text{Prob}(x_i \text{ not evicted}) \\ &= \frac{m}{t-1} \left(1 - \frac{1}{m}\right) \end{aligned}$$

$$\text{Prob}(A) = m/t$$

$$\text{Now: } \frac{m}{t-1} \left(1 - \frac{m}{t}\right) + \frac{m}{t-1} \left(1 - \frac{1}{m}\right) \frac{m}{t} = \frac{m}{t}$$

□

Sampling-based applications: Frequent Items

Problem

Given a stream Σ of n items and a frequency threshold $\varphi \in (0, 1)$, determine all distinct items that occur at least $\varphi \cdot n$ times in Σ (we call them **frequent items**).

Observations

- There can be at most $1/\varphi$ frequent items in Σ .
- Any exact streaming strategy requires space at least linear in the number of distinct elements in Σ (could be arbitrarily close to n)
- We can save space if we give up exactness!

Approximate Frequent Items

Problem

ϵ -Approximate Frequent Items (ϵ -AFI) problem: Given Σ , φ and $\epsilon \in (0, \varphi)$, return a set of distinct items that

- includes all items occurring at least $\varphi \cdot n$ times in Σ .
- contains no item occurring less than $(\varphi - \epsilon) \cdot n$ times in Σ .

Remark: with this formulation, we avoid false negatives but tolerate high-frequency false positives.

Frequent Items with Reservoir sampling

Frequent items can be approximated through **reservoir sampling**:

- 1 Extract an m -sample S of Σ with reservoir sampling (consider all items in Σ as distinct). $\Rightarrow$ each $x \in \Sigma$ is in S with probability $\frac{m}{n}$
- 2 Return the subset $S' \subseteq S$ of distinct items in the sample $n = |\Sigma|$

It can be shown that

let $n_x = \# \text{ occurrences of } x \text{ in } S$
 $x \equiv \text{frequent item} \Rightarrow E[n_x] \geq \varphi \cdot n \cdot \frac{m}{n} \geq 1$

- If $m \geq 1/\varphi$, we expect to find at least one copy of a frequent item in S (and hence in S').
- If we set $m = \lceil 1/\varphi \rceil$ and run $\Theta(\log(1/(\varphi\delta)))$ parallel instances of the algorithm, then S' will contain all frequent items with probability at least $1 - \delta$.

Use Chernoff + Union bound

However: Items with very low frequency (e.g., frequency $1/n$) can still be returned, hence the algorithm does not properly solve the ϵ -AFI problem.